

Kalman Filter-Based Estimation of IMU Measurements

A Bias-Aware Framework for Free Fall and 2D Pendulum Dynamics

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1 Problem Statement and Motivation

1.1 Why IMU State Estimation Matters

Inertial measurement units (IMUs) are compact motion sensors that combine accelerometers and gyroscopes to measure how an object moves over time. Together, these form a six-dimensional IMU measurement vector,

$$\mathbf{z}_k = [a_{x,k} \ a_{y,k} \ a_{z,k} \ \omega_{x,k} \ \omega_{y,k} \ \omega_{z,k}]^\top.$$

IMUs are widely used in drones, smartphones, robotics, and many other systems [1]. However, this advantage comes with an important limitation: raw IMU measurements are never perfectly clean. In practice, the measured signal is a combination of the true motion signal, sensor bias, and random noise, which can be seen as

$$\tilde{\mathbf{z}}_k = \mathbf{z}_k + \mathbf{b}_k + \mathbf{n}_k,$$

where $\tilde{\mathbf{z}}_k$ is the measured IMU signal, $\mathbf{z}_k$ is the true underlying motion signal, $\mathbf{b}_k$ is a slowly varying sensor bias, and $\mathbf{n}_k$ is random measurement noise. As IMU-based motion reconstruction requires integration over time, persistent measurement bias can accumulate into state-estimation error [2].

A basic motion state can be written as

$$\mathbf{x}_k = [\varphi_k \ \theta_k \ \psi_k \ v_{x,k} \ v_{y,k} \ v_{z,k}]^\top.$$

where φ is roll, θ is pitch, and ψ is yaw, describing rotation about the body axes. v_x, v_y, v_z are linear velocities. The central filtering problem is to recover $\mathbf{x}_k$ from corrupted IMU measurements $\tilde{\mathbf{z}}_k$.

1.2 Choice of Controlled Motions

Because IMU measurements contain both acceleration and angular velocity components, we selected two controlled motions that emphasize different parts of the estimation problem: pendulum motion and free fall.

1.2.1 Pendulum Motion

Pendulum motion provides a nonlinear, oscillatory system in which direction changes continuously over time. In the small-angle approximation, the angular displacement can be written as

$$\theta(t) \approx A \sin(\omega t), \quad \dot{\theta}(t) \approx A\omega \cos(\omega t).$$

Thus, the gyroscope signal is strongly time-varying: it is largest near the bottom of the swing and approximately zero near the turning points [3].

In our project, pendulum motion was modeled in the x - z plane. This motion is useful because it stresses orientation tracking. The pendulum repeatedly changes direction, so the filter must track changing angular motion using the gyroscope while also accounting for accelerometer measurements affected by gravity, centripetal acceleration, noise, and bias. Here the key estimation challenge is the nonlinear relationship between angular motion, orientation, and velocity.

1.2.2 Free Fall

Free fall provides a simpler motion model dominated by vertical acceleration. If air resistance is neglected and the vertical direction is chosen as the z -axis, then the ideal position model is

$$z(t) = z_0 + v_0 t - \frac{1}{2}gt^2, \quad v_z(t) = v_0 - gt, \quad a_z(t) = -g,$$

where $g = 9.81 \text{ m/s}^2$ [4]. Free fall is useful because it isolates acceleration-driven motion. In an ideal drop without rotation, the angular velocity should be close to zero, while the vertical acceleration dominates the IMU signal. This makes free fall a useful test case for studying how accelerometer measurements are filtered and integrated into velocity.

2 Madgwick Filter Baseline

We first considered the Madgwick filter as a lightweight baseline for IMU-based orientation estimation. The filter was implemented following Madgwick, 2010 [5]. The filter represents orientation as a quaternion,

$$\mathbf{q}_k = [q_{1,k} \ q_{2,k} \ q_{3,k} \ q_{4,k}]^\top \longrightarrow (\varphi_k, \theta_k, \psi_k),$$

which can later be converted into roll, pitch, and yaw for interpretation.

The Madgwick filter combines two IMU signals in different ways. The gyroscope measurement drives the orientation update through the quaternion derivative,

$$\boldsymbol{\omega}_k = [0 \ \omega_{x,k} \ \omega_{y,k} \ \omega_{z,k}]^\top, \quad \dot{\mathbf{q}}_{\omega,k} = \frac{1}{2}\mathbf{q}_k \otimes \boldsymbol{\omega}_k.$$

The accelerometer is normalized and used as a gravity-direction reference, producing a gradient-descent correction term. The resulting update is

$$\dot{\mathbf{q}}_{\text{est},k} = \dot{\mathbf{q}}_{\omega,k} - \beta \frac{\nabla f}{\|\nabla f\|}, \quad \mathbf{q}_{k+1} = \mathbf{q}_k + \dot{\mathbf{q}}_{\text{est},k} \Delta t, \quad \mathbf{q}_{k+1} \leftarrow \frac{\mathbf{q}_{k+1}}{\|\mathbf{q}_{k+1}\|}.$$

Here, β controls the strength of the accelerometer[5].

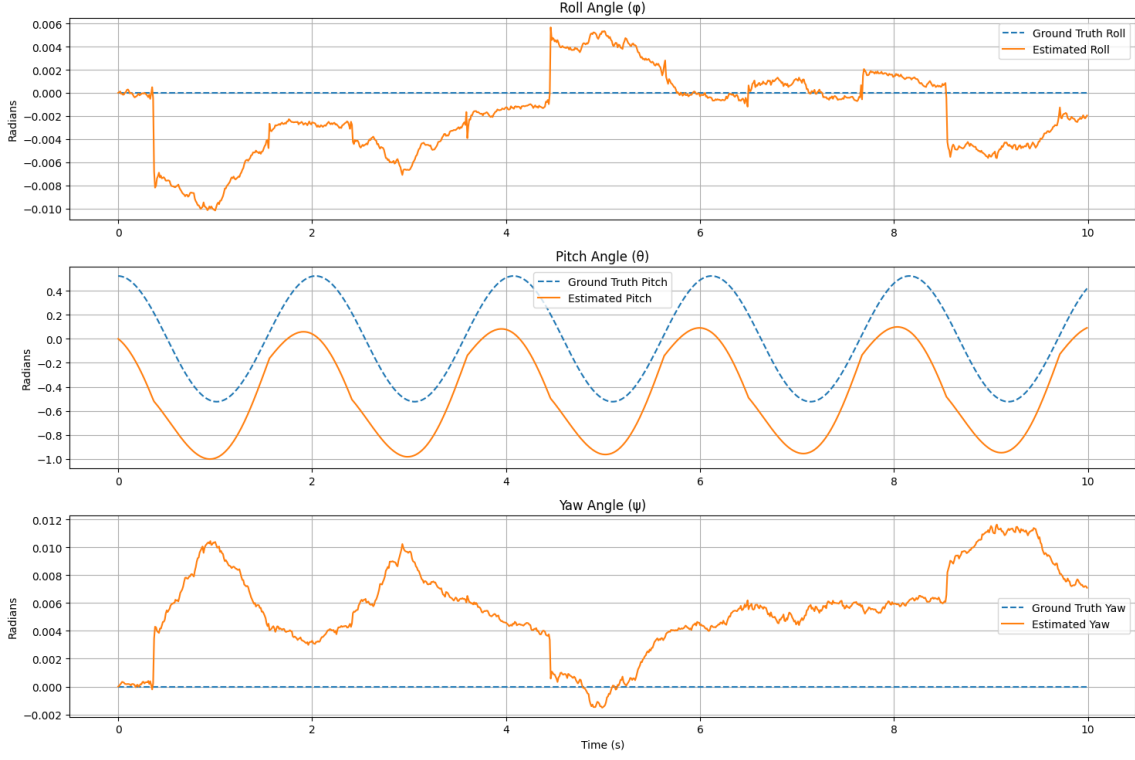


Figure 1: Madgwick orientation estimates for the pendulum simulation. The filter captures the general orientation trend, especially in pitch.

This setup makes Madgwick useful for orientation tracking. However, the accelerometer is used only to correct orientation through the gravity direction, not the translational velocity. This became limiting for our final motion choices as free fall is primarily an acceleration-driven motion based on translational velocity. We therefore shifted the main framework to Kalman filtering and Extended Kalman filtering.

3 Research Questions and Project Contribution

3.1 Research Questions

1. Can an Extended Kalman Filter recover motion states from noisy, biased IMU measurements?
2. Does explicitly estimating sensor bias as part of the state improve performance over treating bias as extra process noise?

3.2 Delta

Our research contribution focuses on the implementation and evaluation of a bias-aware EKF state. The original motion state can be written as an orientation-only state, which can then be expanded to

include translational velocity and further augmented with gyroscope and accelerometer bias terms:

$$\begin{bmatrix} \varphi_k \\ \theta_k \\ \psi_k \end{bmatrix} \longrightarrow \begin{bmatrix} \varphi_k \\ \theta_k \\ \psi_k \\ v_{x,k} \\ v_{y,k} \\ v_{z,k} \end{bmatrix} \longrightarrow \begin{bmatrix} \varphi_k \\ \theta_k \\ \psi_k \\ v_{x,k} \\ v_{y,k} \\ v_{z,k} \\ \mathbf{b}_{g,k} \\ \mathbf{b}_{a,k} \end{bmatrix}.$$

where

$$\mathbf{b}_{g,k} = \begin{bmatrix} b_{g,x,k} \\ b_{g,y,k} \\ b_{g,z,k} \end{bmatrix}, \quad \mathbf{b}_{a,k} = \begin{bmatrix} b_{a,x,k} \\ b_{a,y,k} \\ b_{a,z,k} \end{bmatrix}.$$

The bias terms are modeled as slowly varying random-walk states:

$$\mathbf{b}_{g,k+1} = \mathbf{b}_{g,k} + \boldsymbol{\eta}_{g,k}, \quad \mathbf{b}_{a,k+1} = \mathbf{b}_{a,k} + \boldsymbol{\eta}_{a,k},$$

where $\boldsymbol{\eta}_{g,k}$ and $\boldsymbol{\eta}_{a,k}$ are small process-noise terms (white noise). The augmented EKF estimates the bias terms as their own states. In contrast, a non-augmented model does not correct $\mathbf{b}_g$ or $\mathbf{b}_a$ as states, so persistent bias is absorbed into the motion estimate or treated only as additional uncertainty.

4 Data Generation

4.1 Synthetic Data

Five independent trials were generated per motion type using fixed random seeds {42, 7, 123, 256, 999}, each sampled at 100 Hz ($\Delta t = 0.01$ s).

4.1.1 Synthetic Free Fall

A one-dimensional free-fall scenario was simulated over a total duration of $T_{\text{total}} = 2.0$ s at a sampling rate of 100 Hz ($\Delta t = 0.01$ s), yielding $N = 200$ timesteps per trial. The object remained stationary prior to release at $T_{\text{release}} = 0.3$ s, underwent free fall until impact at $T_{\text{impact}} = 1.5$ s, and was held fixed thereafter.

4.1.2 Synthetic Pendulum

Synthetic IMU data is generated for a simple pendulum swinging in the x - z plane over $N = 1000$ time steps. The pendulum length was set to $L = 1.0$ m with an initial angle of $\theta_0 = \pi/4$ rad and zero initial angular velocity.

4.2 Real Data

For real data, we focused on free fall motion due its feasibility. Data was collected using an Arduino Nano 33 BLE Sense Rev2, which contains an onboard BMI270 6-axis IMU. The BMI270 was configured at its default 100 Hz output rate, with an effective logged rate of approximately 67 Hz due to serial throughput.

Sensor biases were estimated from the first 30 samples of each trial, during which the IMU was held upright, and the biases were subtracted from all subsequent measurements. Release was detected when a_z fell below 0.30 g for three consecutive samples, and impact was detected when a_z exceeded

2 g. Detected free-fall durations ranged from roughly 0.45 to 0.55 s, corresponding to drop heights of approximately 1.0–1.5 m.

We also reconstructed an analytical reference from the detected release time under the ideal free-fall model:

$$z_{true}(t) = \frac{1}{2}g(t - t_{release})^2, v_{z,true}(t) = g(t - t_{release}), a_{z,true}(t) = g$$

5 KF and Extended KF Algorithm for Free Fall and Pendulum Motion

5.1 Process Model

5.1.1 Orientation Update

For pendulum, euler angles are propagated via first-order integration:

$$\begin{pmatrix} \phi_{k+1} \\ \theta_{k+1} \\ \psi_{k+1} \end{pmatrix} = \begin{pmatrix} \phi_k \\ \theta_k \\ \psi_k \end{pmatrix} + \Delta t \cdot T(\phi_k, \theta_k) \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix} + \mathbf{w}_k^{orient}$$

where $T(\phi, \theta)$ is the **Euler angle kinematic matrix**:

$$T(\phi, \theta) = \begin{pmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi \sec \theta & \cos \phi \sec \theta \end{pmatrix}$$

5.1.2 Velocity Update

Velocity for pendulum is updated by integrating the specific force with gravity projected onto the body frame, plus a centripetal correction for the pendulum constraint:

$$\begin{pmatrix} v_{x,k+1} \\ v_{y,k+1} \\ v_{z,k+1} \end{pmatrix} = \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} + \Delta t \begin{pmatrix} \tilde{a}_x - g \sin \theta \\ \tilde{a}_y + g \sin \varphi \cos \theta \\ \tilde{a}_z + L\dot{\theta}_c^2 - g \cos \varphi \cos \theta \end{pmatrix} + \mathbf{w}_k^{vel}$$

Because the accelerometer measures specific force rather than true kinematic acceleration, the gravitational component must be subtracted from the raw reading before the remaining signal can be attributed to motion[2]. The term $L\dot{\theta}_c^2$, where $\dot{\theta}_c = \tilde{\omega}_y - b_{gy}$, is the *centripetal correction*: it accounts for the radial acceleration of the pendulum bob along the z -axis of the body frame. Here, $\tilde{\omega}_y$ is the raw gyroscope measurement along the y -axis, and b_{gy} is its corresponding bias.

5.2 Measurement Model

The measurement model for pendulum stacks the accelerometer and gyroscope predictions into a single vector,

$$h(\mathbf{x}) = h_{true}(\mathbf{x}) + \mathbf{b} = \begin{bmatrix} h_{accel}(\mathbf{x}) \\ h_{gyro}(\mathbf{x}) \end{bmatrix} + \begin{bmatrix} \mathbf{b}_a \\ \mathbf{b}_g \end{bmatrix}, \quad (1)$$

where $h_{true}(\mathbf{x})$ is the noise-free physical prediction and $\mathbf{b}_a, \mathbf{b}_g \in \mathbb{R}^3$ are the accelerometer and gyroscope bias vectors. The biases are included so that the EKF innovation $\mathbf{z}_k - h(\hat{\mathbf{x}}_k)$ compares like with like against the bias-corrupted raw measurements $\mathbf{z}_k$.

5.3 EKF and KF Prediction and Update Step

For free-fall motion, we used the standard linear KF [8]. The measurement model $\mathbf{H}$ is constant in this case since we filtered the accelerometer readings directly. Namely, $\mathbf{H} = [0, 1]$. Considering the linear discrete time kinematics of the state $\mathbf{x} = [v_z, a_z]^T$, the process model $\mathbf{F}$ can be written as

$$\mathbf{F} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}$$

Unlike free fall, the pendulum swing exhibits nonlinear dynamics, so the standard Kalman filter, which assumes fixed linear matrices $\mathbf{F}$ and $\mathbf{H}$, is no longer applicable. We therefore adopt the *Extended Kalman Filter* (EKF), which linearises both the process and measurement models about the current state estimate at each timestep via their respective Jacobians,

$$\mathbf{F}_k = \left. \frac{\partial f}{\partial \mathbf{x}} \right|_{\hat{\mathbf{x}}_k}, \quad \mathbf{H}_k = \left. \frac{\partial h}{\partial \mathbf{x}} \right|_{\hat{\mathbf{x}}_k^-}, \quad (2)$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ maps the state $\mathbf{x}_k$ to its prediction $\mathbf{x}_{k+1}$ discussed in Sections 5.1, and h is the nonlinear measurement model of Section 5.2. Here, the superscript $-$ denotes the prior state estimate $\hat{\mathbf{x}}_k^-$, propagated forward by the process model but not yet corrected by the incoming measurement. The prediction and update steps then follow the standard EKF algorithm [8].

6 Results

6.1 Synthetic Free-Fall Simulation

We first tested the Kalman Filter on synthetic free-fall data, where the ground truth is known from the constant-acceleration model. The synthetic setup is summarized in Table 1. For this case, the state was restricted to vertical velocity and acceleration:

$$\mathbf{x}_k = [v_{z,k} \quad a_{z,k}]^T.$$

Position was not included as a state; instead, it was computed afterward by integrating the filtered velocity estimate.

Table 1: Synthetic free-fall simulation setup.

Quantity	Value
Number of trials	5 random seeds
Free-fall start time	$t = 0.3$ s
Impact time	$t = 1.5$ s
Theoretical height	$\frac{1}{2}g(1.2)^2 \approx 7.1$ m
State vector	$\mathbf{x}_k = [v_{z,k}, a_{z,k}]^T$
Metrics	RMSE and maximum error

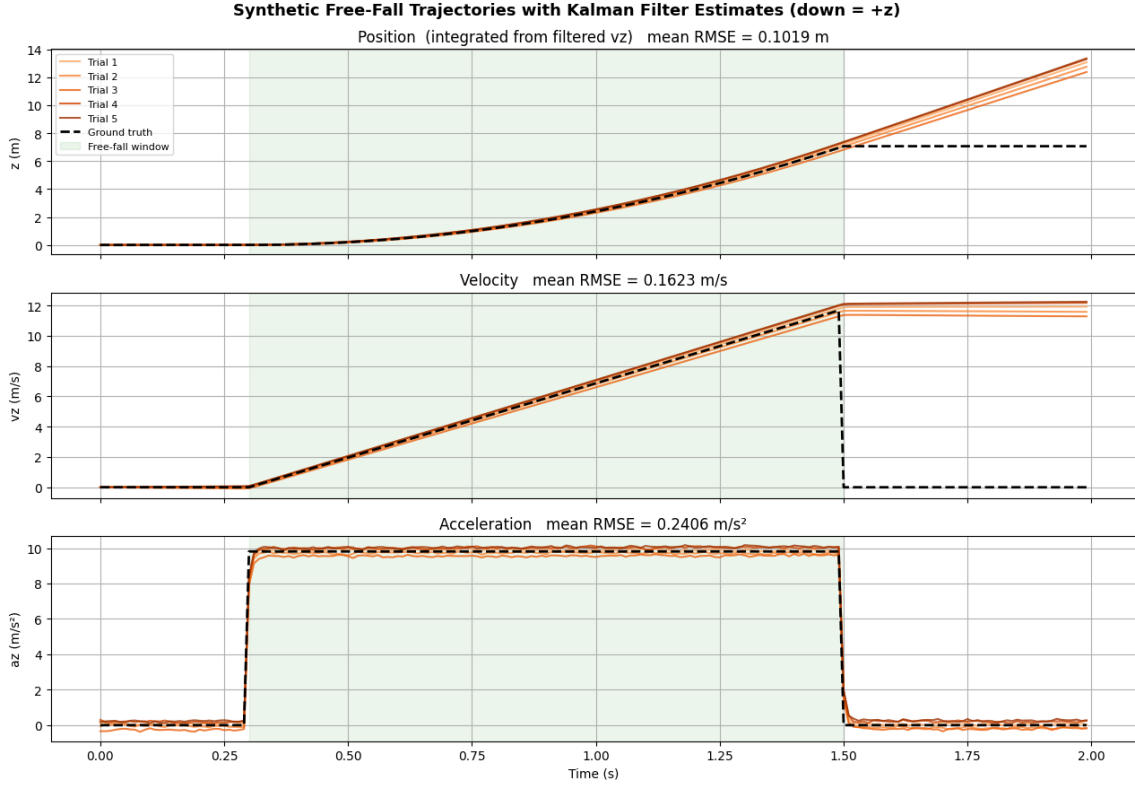


Figure 2: Synthetic free-fall Kalman Filter results. Position is integrated from the filtered velocity estimate.

Across the five trials, the Kalman Filter tracked the synthetic velocity and acceleration trends with low error. The integrated position estimate also remained close to the theoretical drop height, showing that the filter performs well when the data match the assumed linear free-fall model.

6.2 Real Free-Fall Data

We collected IMU measurements using an Arduino Nano 33 BLE board with an embedded IMU. The board was manually held at rest and then dropped onto a sponge pad (Figure 3).



Figure 3: Real free-fall experiment setup using an Arduino Nano 33 BLE board with an embedded IMU.

The shaded free-fall window corresponds to the interval after release and before impact, where the free-fall model is most physically appropriate (Figure 4).

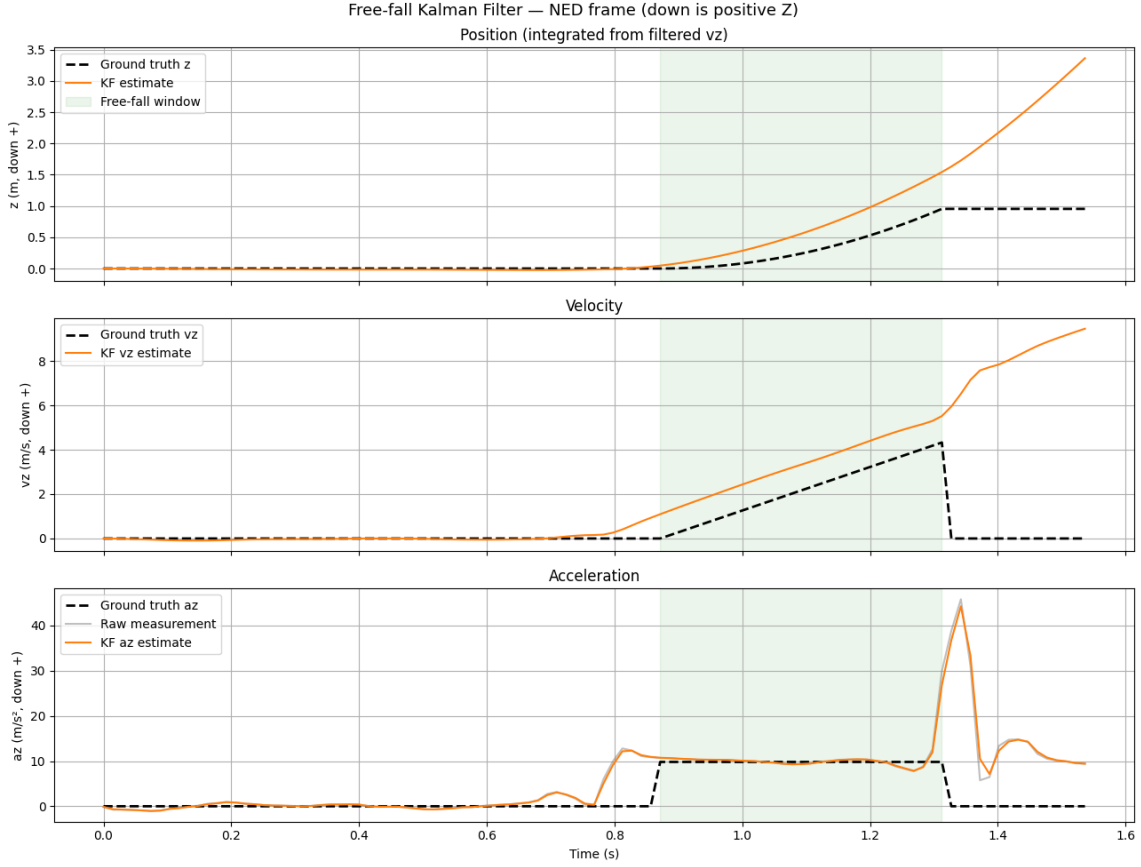


Figure 4: Real free-fall Kalman Filter results. The green region marks the free-fall window used for RMSE evaluation.

During the free-fall window, the filtered acceleration followed the expected gravity-dominated regime, and the filtered velocity captured the increasing downward trend. Position showed larger divergence because it was obtained by integrating the filtered velocity rather than directly measured, where error would have accumulated over time (Figure 4).

The RMSE values over the free-fall window were $\text{RMSE}_z = 0.356$ m, $\text{RMSE}_{v_z} = 1.156$ m/s, $\text{RMSE}_{a_z} = 3.146$ m/s². These values show that the filter recovered the broad free-fall behavior, but real IMU data remained harder than synthetic data because of noise, timing alignment, and impact dynamics - exactly what makes real data crucial to stress-testing.

6.3 Synthetic Pendulum Simulation

For the pendulum case, we used synthetic data so that the true trajectory was known. The pendulum was modeled in the x - z plane with no lateral motion, making v_y expected to remain near zero.

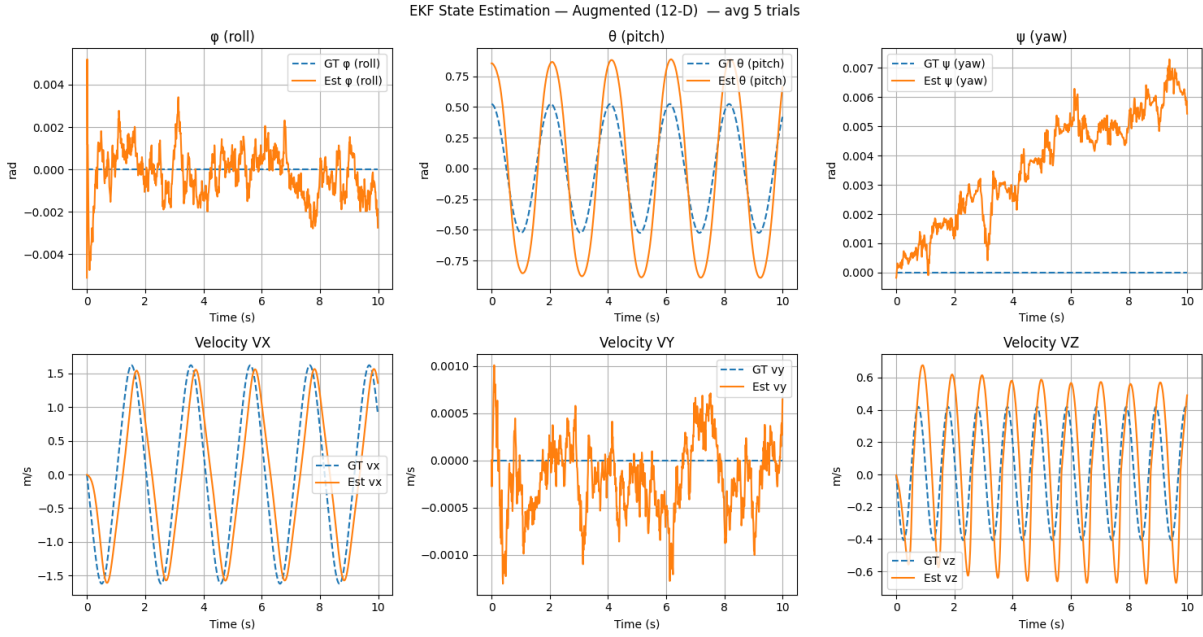


Figure 5: Synthetic pendulum EKF results comparing estimated and ground-truth velocities and Euler angles.

The EKF captured the main pendulum structure: v_x and v_z followed the swing dynamics, while v_y stayed close to zero as expected from the planar motion assumption. Roll and yaw also remained near zero, while pitch carried the dominant oscillatory orientation signal (Figure 5).

STATE	UNIT	NON-AUG	AUG	Δ RMSE	NON-AUG	AUG	Δ MAX
RMSE					MAX ERROR		
VELOCITIES							
vx	m/s	0.8070	0.5901	-26.9%	1.1541	0.9186	-20.4%
vz	m/s	0.6162	0.3212	-47.9%	1.0669	0.5856	-45.1%
vy	m/s	0.0018	0.0005	-55.6%	0.0047	0.0014	-63.8%
ANGLES							
ψ (yaw)	rad	0.0027	0.0022	-18.5%	0.0054	0.0044	-18.5%
θ (pitch)	rad	0.2624	0.2915	+11.1%	0.4008	0.3920	-2.2%
ϕ (roll)	rad	0.0009	0.0012	+33.3%	0.0055	0.0056	+0.0%

Figure 6: Comparison of augmented and non-augmented EKF performance for synthetic pendulum data.

Bias augmentation improved the velocity estimates most clearly. In particular, the augmented EKF reduced RMSE for v_x and v_z by approximately 27% and 48%, respectively. This supports the idea that explicitly estimating bias can improve acceleration-driven velocity recovery. For orientation,

the effect was more mixed: pitch RMSE became slightly worse, while maximum error improved. This suggests that augmentation can reduce worst-case drift but may also introduce additional sensitivity (Figure 6).

6.4 Overall Comparison

Table 2 summarizes the main results across the two motion systems.

Table 2: Summary of filtering choices and main results.

	Pendulum	Free Fall
Filtering algorithm	Extended Kalman Filter	Kalman Filter
Dynamics	Nonlinear, oscillatory motion	Approximately linear vertical motion
State vector	$\mathbf{x}^{\text{nonaug}} = [\varphi, \theta, \psi, v_x, v_y, v_z]^\top$; $\mathbf{x}^{\text{aug}} = [\varphi, \theta, \psi, v_x, v_y, v_z, \mathbf{b}_g, \mathbf{b}_a]^\top$	$\mathbf{x} = [v_z, a_z]^\top$; position integrated from filtered velocity
Main result	Bias augmentation improved velocity RMSE, especially v_x and v_z , but had mixed effects on orientation RMSE	Filter tracked acceleration and velocity during the free-fall window; position error accumulated through integration

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